Soumya Kanti Upadhyay

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EDUCATION

Jalpaiguri Government Engineering College **2024 – 2028**
B.Tech - Computer Science and Engineering - ***CGPA - 9.9*** *Jalpaiguri, West Bengal*

Baharu High School **2020 – 2022**
12th - WBCHSE Board - ***Percentage - 82.4%*** *Kolkata, West Bengal*

EXPERIENCE

Research Intern — National Institute of Technology (NIT) Durgapur **June 2026 - July 2026**
Multimodal Colon Cancer Detection Model *Durgapur, West Bengal*

- Developed a **confidence-aware multimodal deep learning framework** integrating colonoscopy videos, histopathology images, and IHC biomarkers for colorectal polyp prognosis.
- Implemented **VideoMAE and DINOv2-based feature extraction** with temporal, patch-level, and patient-level attention mechanisms using PyTorch.
- Applied **5-fold Stratified Cross-Validation, AdamW, CosineAnnealingLR, and early stopping** for robust model training and evaluation.
- Evaluated model performance using **Accuracy, Precision, Recall, Macro F1, ROC-AUC, Sensitivity, Specificity, ECE, Brier Score, and Confusion Matrix**.

PROJECTS

Snap Class | Python, Streamlit, Dlib, Resemblyzer, Librosa, Scikit-learn, Supabase, Flask **May 2026**

- Developed an AI-powered attendance system using **facial recognition and voice biometrics** to automate student identification and attendance tracking.
- Implemented **Face ID and Voice ID pipelines** for biometric embedding generation and matching, enabling fast and reliable attendance marking.
- Built interactive **teacher and student dashboards** with course management, QR-based enrollment, biometric registration, attendance marking, and attendance history.
- Integrated **Supabase PostgreSQL** for managing student, teacher, subject, biometric, enrollment, and attendance records, with a Flask-based landing page.

Real Time AI GYM Trainer | Python, Streamlit, MediaPipe, OpenCV, Groq, gTTS **August 2026**

- Developed a real-time AI fitness coach using **MediaPipe Pose Landmarker** and computer vision to detect body posture and analyze exercise form.
- Implemented real-time **rep counting, exercise tracking, and form correction** for squats, push-ups, biceps curls, shoulder presses, and lunges.
- Integrated **LLM-powered coaching and text-to-speech feedback** using Groq and gTTS to provide personalized guidance during workouts.
- Built an interactive **Streamlit web application** with webcam/video processing for real-time exercise monitoring and performance feedback.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, HTML, CSS, JavaScript, React, SQL.
Technologies/Frameworks: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, Flask, Streamlit.
Developer Tools: VS Code, Jupyter Notebook, MySQL, PyCharm, Canva, Excel.
Core Concepts: Data Structures and Algorithms, Artificial Intelligence, Software Engineering, Operating Systems, DBMS, OOPS.

PROBLEM SOLVING

- Have solved 1000+ problems on Leetcode, Max Rating : 2087(Knight).
- Have solved 300+ problems on CodeChef, Max Rating : 1667.
- Have solved 200+ problems across various platforms like GeeksforGeeks, Algo University.

ACHIEVEMENTS

- Qualified for Round 2 of Naukri Campus Young Turks 2026, scoring **96.85 percentile** in Round 1.
- Participated in **AINCAT 2026**, conducted by Naukri Campus.
- Completed Dynamic Programming and Graph Theory Camps by AlgoUniversity.